

Article

Geopolitical Risk and Shipping Supply Chain Resilience: Systemic Characteristics, Impact Mechanisms, and the Security of Logistics Nodes

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Abstract

Understanding how geopolitical risk propagates through shipping networks to impact shipping supply chain resilience (SSCR) is essential for advancing global maritime governance reform. This study examines the systemic effects of geopolitical risk on SSCR using cross-border panel data derived from international shipping networks and identifies the transmission mechanisms operating through critical logistics nodes. The results indicate that geopolitical risk exerts a significant and persistent negative impact on SSCR, with significant multidimensional heterogeneity. Mechanism analysis shows that SSCR is undermined through three channels: logistics infrastructure disruption, increased freight rate volatility, and reduced customs clearance efficiency. Node-level evidence further reveals consistently negative effects across most critical logistics nodes. Logistics infrastructure disruption is particularly pronounced in ports. Logistics nodes along Indian Ocean routes exhibit more pervasive effects through the freight rate volatility channel, while reduced customs clearance efficiency represents a common transmission channel across most nodes.

Keywords: geopolitical risk; shipping supply chain resilience; logistics nodes

1. Introduction and Literature Review

In recent years, global instability has intensified, with a growing number of geopolitical flashpoints emerging worldwide. The Russia–Ukraine conflict has turned the Black Sea into a militarized zone; the Red Sea crisis and Houthi attacks have disrupted traffic along the Suez Canal route; and escalating tensions between the United States and Iran have increased transportation risks in the Strait of Hormuz [1–3]. Alongside persistent geopolitical tensions, including disputes in the South China Sea and the Israel–Palestine conflict, these developments have heightened uncertainty in cargo sourcing, increased the complexity of shipping routes, and amplified freight rate volatility [4,5]. Against this increasingly complex international landscape, a rigorous assessment of how geopolitical risk affects shipping supply chain resilience (SSCR) has become an important research agenda for advancing global maritime governance reform.

Research on SSCR has attracted increasing attention. The literature mainly focuses on two aspects. The first concerns the conceptualization of SSCR. Supply chain resilience is commonly defined as a dynamic process spanning the pre-shock, shock, and post-shock stages. It includes anticipatory capabilities such as prediction, redundancy, and network diversification before disruptions; absorptive and response capacities during



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disruptions; and recovery and adaptation capabilities after disruptions [6,7]. Empirically, this definition often relies on transaction- or firm-level high-frequency data, which may be subject to limited observability. Some studies adopt a regional or national perspective and extend supply chain resilience to a country's ability to maintain stable operations under external shocks [8,9]. From this perspective, resilience extends beyond firm-level recovery performance and reflects a forward-looking capacity for risk mitigation and systemic adaptation [5,10]. Building on this literature, this study defines SSCR as a country's ability, under external uncertainties such as geopolitical shocks, to reduce disruption risks and maintain stable operations through the optimization of shipping network structures, institutional arrangements, and resource allocation. The openness of shipping networks and their strong cross-jurisdictional dependence further distinguish SSCR from resilience frameworks developed for manufacturing supply chains [11,12]. These differences are particularly evident across functional, structural, and governance dimensions [13–19] (Appendix A).

The second strand of the literature focuses on the measurement of SSCR and the analysis of resilience-enhancing strategies. Existing approaches mainly include network-based models, reliability and optimization models, and multi-criteria decision-making methods. These methods quantify vulnerability, recovery, and key resilience drivers through simulation, optimization, and expert-based evaluation, thereby supporting empirical analysis and enabling cross-case comparisons [11,12,20–23] (Appendix B). The strategies for enhancing SSCR largely mirror those developed for general supply chains. They are typically categorized into proactive strategies, which aim to prevent or prepare for disruptions, and reactive strategies, which focus on post-disruption recovery [24,25]. While these approaches help reduce vulnerability and improve recovery capacity, they often assume relatively stable institutional environments. As a result, they tend to understate the geopolitical sensitivity and spatial concentration of risks at key shipping nodes, limiting their applicability in highly politicized maritime contexts.

As sources of global instability and geopolitical flashpoints continue to proliferate, growing attention has been paid to the impact of geopolitical risk on supply chain resilience through cascading disruptions [26]. Specifically, geopolitical risk can trigger functional disruptions at critical logistics nodes, generating cascading effects across supply networks [27]. Scholars argue that these impacts are concentrated at critical logistics nodes, which exhibit multiple layers of vulnerability. These include physical fragility due to aging infrastructure, institutional fragility arising from overlapping national interests, and cyber vulnerability associated with exposed digital control systems. Such characteristics make these nodes focal points of risk transmission [28,29]. Although technologies such as blockchain can provide more transparent and secure routing alternatives, geopolitically induced disruptions still necessitate supply chain reconfiguration. This process not only increases operational costs but also introduces additional uncertainty and vulnerability [30]. Moreover, frequent reconfiguration prevents supply chains from achieving stable operational equilibria, thereby undermining resilience and responsiveness [31]. Finally, blockades or the militarization of supply chain nodes not only directly sever maritime corridors but also generate cascading disruptions, including cargo congestion and capacity mismatches, which may escalate into systemic congestion across the global maritime transport system [26,32].

With the intensification of geopolitical competition, countries increasingly resort to asymmetric policy instruments such as tariff escalation, export controls on advanced technologies, and entity list sanctions [33]. These measures raise compliance costs, compress profit margins, and redirect trade flows, thereby forcing multinational firms to decouple and regionally reconfigure their supply chains, ultimately leading to a more fragmented and bloc-based global supply chain structure [13]. Some studies further argue that com-

plex international conditions and geopolitical tensions prompt firms to adopt big data analytics to anticipate the impacts of geopolitical shocks and develop proactive contingency plans [24]. Such adaptive responses, supported by continuous innovation, enhance the intelligence, efficiency, and resilience of supply chain systems.

The above review shows that a substantial body of the literature has examined the impact of geopolitical risk on supply chain resilience, providing a foundation for this study. However, existing research largely focuses on manufacturing or general supply chain frameworks, with limited attention to shipping supply chains. In particular, the mechanisms through which geopolitical risk affects SSCR remain underexplored. Due to their globally networked structure, dependence on specific maritime corridors, and high asset specificity, shipping supply chains exhibit distinct risk transmission mechanisms and resilience response logics compared with general supply chain frameworks. Ignoring this sector-specific context may lead to overly generalized resilience strategies. Moreover, critical logistics nodes serve as both strategic hubs and systemic bottlenecks within shipping supply chains. Yet, existing studies rarely assess how geopolitical risk at these nodes affect SSCR. This gap may obscure node-specific vulnerabilities and create blind spots in risk early warning systems, thereby underestimating potential threats to global economic security. Against this backdrop, how does geopolitical risk affect SSCR, and through which mechanisms does this effect operate? How do these effects vary across critical logistics nodes? These questions remain insufficiently explored and warrant systematic investigation.

Building on this, this study employs cross-border panel data on shipping from 2007 to 2023 to examine the impact of geopolitical risk on SSCR and its underlying mechanisms, and further examines its effects at critical logistics nodes. Compared with the existing literature, this study makes the following three main contributions.

First, it moves beyond the path dependence of prior research that primarily focuses on manufacturing or general supply chain frameworks. From a country-level perspective, it conceptualizes SSCR as a systemic capability embedded in shipping network structures, macro-institutional environments, and resource allocation processes. Based on this definition, the study develops a three-dimensional evaluation framework of SSCR—encompassing functional, structural, and governance dimensions—and empirically examines how geopolitical risk affects SSCR and its transmission mechanisms. This framework provides a basis for developing proactive and adaptive resilience governance in the shipping sector.

Second, to better capture the sector-specific nature of geopolitical risk, this study adapts the general geopolitical risk measurement framework proposed by Caldara and Iacoviello [4] by incorporating a shipping-specific lexicon. It introduces additional dimensions related to shipping activities, firm operations, and port and shipping infrastructure. By constructing a three-tier indicator system—“terms–topics–behaviors”—the study addresses the limitations of the original framework in identifying shipping-related risks, thereby extending the measurement of geopolitical risk from a general framework to a sector-specific analytical tool.

Third, departing from the conventional treatment of supply chains as homogeneous networks that overlook spatial heterogeneity, this study’s extended analysis focuses on three major strategic maritime corridors—the Pacific route, the Indian Ocean route, and the Atlantic route. In this context, the effects of geopolitical risk originating from critical logistics nodes on SSCR are systematically examined. This enhances the spatial dimension of SSCR analysis and provides a new paradigm for designing risk containment mechanisms for shipping supply chains anchored in critical logistics nodes.

The remainder of this study is structured as follows. Section 2 develops the theoretical mechanisms and research hypotheses. Section 3 describes the empirical methodology and

data. Section 4 presents the empirical results. Section 5 discusses the main findings and their implications. Section 6 concludes the study.

2. Theoretical Mechanisms and Hypotheses

2.1. Mechanism of Logistics Infrastructure Disruption

Geopolitical risk undermines SSCR by disrupting logistics infrastructure. From the perspective of institutional economics, geopolitical risk can be interpreted as uncertainty embedded in the institutional environment. When formal rules ensuring the smooth functioning of logistics systems are eroded or break down, interactions among actors may shift from rule-based cooperation to conflict driven by coercive forces. Under such conditions, logistics infrastructure becomes highly vulnerable to direct physical damage [5,14]. Moreover, from the perspective of resource allocation theory, geopolitical risk further undermines SSCR by distorting resource distribution. Amid heightened security threats, ports and shipping firms are often forced to divert substantial resources toward security enhancement and risk mitigation. This reallocation generates inefficiencies and crowds out investment in infrastructure maintenance and upgrading. Meanwhile, economic sanctions may disrupt the supply of critical equipment and materials, thereby constraining the maintenance, renewal, and normal operation of port and shipping infrastructure, exacerbating damage to the infrastructure [21,27].

Path dependence theory suggests that once established, logistics networks tend to evolve along relatively stable trajectories. When infrastructure functionality deteriorates due to shocks, supply chains may become locked into inefficient operational patterns, resulting in reduced flexibility and adaptability [31]. In such contexts, response times are prolonged, and the capacity to adjust transport planning and logistics arrangements in a timely manner is significantly reduced. At the same time, the availability of alternative infrastructure—such as backup ports and substitute berths—declines, further limiting feasible responses to disruptions. Consequently, SSCR is undermined as supply chains struggle to effectively absorb shocks and adapt to changing conditions. Accordingly, the following hypothesis is proposed:

Hypothesis 1: *Geopolitical risk undermines SSCR by disrupting logistics infrastructure.*

2.2. Mechanism of Increased Freight Rate Volatility

Geopolitical risk undermines SSCR by increasing the volatility of shipping freight rates. Location theory posits that geographic characteristics shape transportation costs, which in turn influence trade flows and the spatial configuration of global supply chains. The global maritime system relies heavily on critical chokepoints such as the Strait of Hormuz and the Suez Canal. When geopolitical risk threatens these routes, shipping companies may suspend transit or reroute vessels, while incorporating risk premiums into freight rates, thereby amplifying rate volatility [34]. For instance, following the escalation of U.S.–Iran tensions and the associated risks in the Strait of Hormuz, CMA CGM announced an emergency conflict surcharge of USD 2000/3000 per TEU/FEU for dry containers and USD 4000 for reefer containers or special equipment, while Hapag-Lloyd introduced a war risk surcharge of USD 1500 per TEU for standard containers and USD 3500 per TEU for reefer containers and special equipment. Geopolitical risk may also disrupt energy supply, as export restrictions or fluctuations in bunker fuel prices are transmitted to freight rates through cost pass-through mechanisms, further increasing volatility [35].

The heightened volatility in shipping freight rates, in turn, undermines SSCR, as large and frequent price fluctuations make it difficult for firms to accurately forecast and control transportation costs. From the perspective of resource allocation theory, financial capital is

a scarce and constrained resource. When freight rates surge unpredictably, firms may be forced to reallocate funds from other critical budget items to cover rising transportation costs, thereby tightening liquidity conditions [36]. Such financial pressure constrains firms' ability to mobilize resources in response to shocks, reduces operational stability, and ultimately erodes SSCR. Accordingly, this study proposes the following hypothesis:

Hypothesis 2: *Geopolitical risk undermines SSCR by increasing the volatility of shipping freight rates.*

2.3. Mechanism of Reduced Customs Clearance Efficiency

Geopolitical risk undermines SSCR by reducing customs clearance efficiency. From the perspective of institutional economics, property rights protection and contract enforcement constitute the foundation of market functioning. Heightened geopolitical tensions—particularly interstate political conflicts—tend to foster protectionist policies. To safeguard economic security and political interests, governments often adopt restrictive trade measures such as tariff increases, quota restrictions, or embargoes [33]. These policies have increased the compliance costs and uncertainty of trade activities, forcing enterprises to navigate more complex cross-border procedures. At the same time, in the context of institutional instability, information asymmetry or regulatory ambiguity may lead to the delay or detention of goods. These factors collectively reduce the efficiency of customs clearance.

The deterioration of customs clearance efficiency not only prolongs cargo dwell time at ports—thereby disrupting downstream logistics processes and extending overall supply chain response time [5]—but also impairs information transparency. This exacerbates informational asymmetry among upstream and downstream actors in shipping supply chains. According to information asymmetry theory, when one party holds a significant informational advantage, coordination efficiency declines and the risk of market failure increases [37]. Such imbalances constrain firms' ability to respond effectively to demand fluctuations and external shocks, ultimately undermining SSCR. Accordingly, this study proposes the following hypothesis:

Hypothesis 3: *Geopolitical risk undermines SSCR by reducing customs clearance efficiency.*

3. Methodology and Data

3.1. Empirical Model

This study constructs a fixed-effects model to examine the impact of geopolitical risk on SSCR. The model accounts for unobserved individual heterogeneity by allowing for unit-specific intercepts, which capture all the time-invariant characteristics across cross-sectional units. The model assumes that these individual effects are potentially correlated with the explanatory variables. Accordingly, cross-sectional heterogeneity is controlled through unit-specific fixed effects. The following baseline regression model is specified:

$$SSCR_{it} = \zeta_1 SGPR_t + \zeta_2 Z_{it} + \zeta_0 + \delta + \varepsilon_{it} \quad (1)$$

where $SSCR_{it}$ denotes the SSCR of country i in year t , and $SGPR_t$ represents the geopolitical risk in the year t . In addition, fixed effects (δ) are included to control for time-invariant unobserved heterogeneity. ζ_0 is the constant term, ζ_1 and ζ_2 are the estimated coefficients, and ε_{it} denotes the error term.

The vector Z_{it} includes a set of control variables. First, the gross domestic product (GDP) captures a country's economic scale. It directly affects shipping demand and resource allocation capacity and thus constitutes the economic foundation of supply chain

resilience. Second, patent applications (*Patent*) measure technological innovation. A stronger technological capability enhances supply chain efficiency and risk mitigation capacity, thereby improving resilience. Third, the international production position index (*Position*), calculated based on forward production length, reflects a country's role in the global division of labor. It influences dependency and vulnerability within global supply chains, as countries positioned at different stages of production may experience distinct mechanisms through which geopolitical risk is transmitted. Fourth, foreign direct investment stock (*FDI*) captures the sensitivity of cross-border capital flows, which may amplify supply chain vulnerability when exposed to external shocks. Fifth, labor (*Labor*) represents the scale and quality of the workforce. Labor conditions affect production capacity and supply chain flexibility.

3.2. Variables

3.2.1. Dependent Variable: SSCR

This study constructs a measurement framework for SSCR from three dimensions—function, structure, and governance. The functional dimension focuses on the efficiency of spatiotemporal transformation in international cargo transport. Greater origin and destination diversification in the shipping supply chain enhances market coverage and reduces exposure to localized disruptions [38]. Furthermore, against the backdrop of continuously fluctuating shipping freight rates, the ability to arrange international freight in a flexible, reliable, and cost-effective manner is an important indicator of a country's SSCR functional dimension. Therefore, this study uses origin diversification, destination diversification, and the capacity to arrange international freight as indicators of the SSCR functional dimension.

The structural dimension reflects the stability of shipping networks supported by maritime digital technologies and hub-oriented connectivity. The Liner Shipping Connectivity Index (LSCI) measures the density and control of global shipping networks, while the degree of digital transformation in the shipping industry captures the adoption of maritime digital technologies. The value output capability is a direct manifestation of the “value radiation” of a hub-and-spoke network, and it serves as a supplementary measure of the stability and influence of the supply chain structure from the perspective of value output. Together, these indicators jointly characterize the structural resilience of shipping supply chains [39,40].

The governance dimension emphasizes redundancy-oriented resource allocation to mitigate transportation delay risks. Port throughput capacity reflects infrastructural redundancy, while shipping resource allocation efficiency captures the effectiveness of resource deployment under uncertainty [41]. In addition, the higher a country's position in the global production network, the greater its bargaining power in resource allocation and the stronger its capacity to invest in and optimize redundant resources. Therefore, this study uses the above indicator to characterize the governance dimension.

The indicators adopted in this study are closely linked to recent geopolitical risk, which has significantly affected the functional, structural, and governance dimensions of SSCR. The 2022 Russia–Ukraine conflict disrupted global shipping systems, as major carriers suspended services to and from Russia, and Black Sea port operations were constrained, leading to a decline in regional liner shipping connectivity [2]. At the same time, increased geopolitical fragmentation intensified uncertainty in global value chains, adversely affecting both value output capacity and countries' positions in international production fragmentation [5]. The 2023 Red Sea crisis further illustrates how geopolitical shocks reshape shipping performance indicators. Vessel rerouting via the Cape of Good Hope increased transit times by approximately 30% and reduced effective capacity by about 9%, reflecting a decline in supply-side competitive capacity and affecting the ability

to arrange international freight [42]. Meanwhile, reductions in port calls and deployed capacity, along with weakened feeder connectivity, constrained origin–destination diversification and accelerated the reconfiguration of hub networks [43,44]. Increased congestion in Northern European ports further strained the port throughput capacity. In response, firms and countries developed alternative corridors and adopted multi-node strategies, leading to adjustments in resource allocation efficiency and positions in global production networks [45]. Concurrently, digital transformation has enhanced coordination and information transparency, partially strengthening structural resilience [46].

Based on the above framework, Table 1 presents the indicator system used to construct the SSCR index. The entropy method is used to construct the composite SSCR index.

Table 1. Construction of the SSCR Index.

Variable	Dimension	Indicator	Measurement	Direction
Shipping Supply Chain Resilience	Function	Origin diversification of shipping supply chain	Sum of squared shares of shipping value-added imports from each partner country in total shipping value-added imports	—
		Destination diversification of shipping supply chain	Sum of squared shares of shipping value-added exports to each partner country in total shipping value-added exports	—
		Capacity to arrange international freight	Logistics Performance Index: Ease of arranging competitively priced shipments	+
	Structure	Liner shipping connectivity	Liner Shipping Connectivity Index (LSCI) ¹	+
		Degree of digital transformation in the shipping industry	Share of value-added inputs from digital industries in total value-added of the shipping sector	+
		Value output capacity	Shipping industry forward participation index in global value chains	+
		Port throughput capacity	Container throughput of ports	+
	Governance	Shipping resource allocation efficiency	New revealed comparative advantage (RCA) index of the shipping industry ²	+
		Position in the global production network	Position index in the global production network of the shipping industry	+

¹ The LSCI, published by the United Nations Conference on Trade and Development (UNCTAD), measures a country's integration into the global liner shipping network. A higher value indicates better connectivity. ² The traditional RCA index neglects the fact that a sector's value added may be embodied in exports of other domestic sectors through indirect export channels [47]. The New RCA index provides a more accurate measure of the shipping industry's actual contribution to and competitiveness within global value chains, thereby capturing the effectiveness of resource allocation. This study calculates the New RCA index following the method proposed by Wang et al. [48].

3.2.2. Explanatory Variable: Geopolitical Risk

To measure geopolitical risk, this study adopts a hub–country risk perception perspective and extends the general geopolitical risk framework proposed by Caldara and Iacoviello [4] to incorporate the industry-specific characteristics of shipping supply chains. This extension ensures that the constructed index captures variations in geopolitical risk that are directly relevant to shipping activities. China is selected as the reference country for risk perception. According to data from the UNCTADstat Data Center, China is the

world's largest merchandise trading nation, accounting for 14.36% of global trade in 2025. It also ranks first in the LSCI, maintaining this position for 20 consecutive years up to 2025, and accounts for 15.36% of global seaborne trade. Given this central position in both global trade and shipping networks, China's perception of geopolitical risk can generate systemic spillovers through route adjustments, capacity deployment, and port-shipping resource reallocation, thereby affecting the global shipping network. Accordingly, this study draws on authoritative Chinese policy-oriented and industry newspapers to capture China-perceived shipping-related geopolitical risk.

Specifically, fourteen major national newspapers are selected as data sources. Following the iterative keyword identification approach proposed by Davarzani et al. [49], a structured keyword retrieval framework is constructed (Appendix C). Initial geopolitical risk keywords are drawn from the geopolitical risk dictionary developed by Caldara and Iacoviello [4]. Nuclear-related terms are excluded, as nuclear events are largely concentrated in earlier decades and exhibit limited relevance during the sample period. Based on the definition of shipping supply chains, industry- and firm-related shipping keywords are added [14]. Given the critical role of infrastructure in ensuring shipping continuity, port and shipping infrastructure terms are also incorporated [21].

Using this framework, 792 news articles are initially retrieved. CiteSpace 6.3.1 is then applied to conduct keyword co-occurrence network analysis, and high-frequency and burst keywords are incorporated to refine the retrieval structure. After manual screening to remove irrelevant articles, 1499 valid news items are retained. The annual geopolitical risk index is constructed as the ratio of shipping-related geopolitical risk news to total news published in each year and is subsequently normalized for cross-year comparability. A higher value indicates greater geopolitical risk faced by shipping supply chains.

3.2.3. Data

To examine the impact of geopolitical risk on SSCR, this study constructs a panel dataset covering 42 coastal countries over the period 2007–2023. The data are primarily drawn from the Asian Development Bank Multi-Regional Input–Output database (ADB–MRIO), the World Bank database, and the China National Knowledge Infrastructure (CNKI). The dependent variable, SSCR, is mainly derived from the ADB–MRIO database. To ensure data quality and consistency in indicator construction, the original input–output tables are systematically processed and harmonized. Following the gross trade accounting framework proposed by Koopman et al. [50] and Wang et al. [51,52], the ADB–MRIO tables are decomposed to construct a value-added flow matrix. On this basis, indicators are computed, including the origin and destination diversification of shipping supply chains, as well as the degree of digital transformation in the shipping industry. Additional SSCR-related indicators are obtained from the World Bank, UNCTAD, and the Clarksons Research database.

The key explanatory variable, geopolitical risk, is sourced from the CNKI database, while other variables are collected from the World Bank, Clarkson, and UNCTAD databases. After constructing the variables and composite indicators, a series of data quality checks were performed. Country–year observations with missing or abnormal values in key variables are excluded, resulting in a final sample of 714 observations. The variance inflation factor test indicates that multicollinearity is not a serious concern. Through this process, the study constructs a highly reliable panel dataset, thereby enhancing the credibility and replicability of the analysis and providing a solid empirical foundation for subsequent econometric analyses.

4. Empirical Results

4.1. Baseline Regression Results

Table 2 Column (1) presents the fixed-effects (FE) estimates, while Column (2) reports the random-effects (RE) results. The Hausman test strongly rejects the random-effects specification, favoring the fixed-effects model. The results show that the coefficient of geopolitical risk (SGPR) is negative, indicating that higher geopolitical risk leads to a reduction in SSCR.

Table 2. Baseline regression and robustness test results. Robust standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
SGPR	−0.764 ** (0.313)	−0.755 ** (0.305)	−0.455 *** (0.166)	−0.065 *** (0.019)	−0.932 ** (0.367)	−0.016 * (0.009)	−0.681 *** (0.242)
Controls	YES	YES	YES	YES	YES	YES	YES
Constant	−32.943 * (17.944)	−30.286 ** (14.487)	−58.582 *** (15.648)	−41.458 *** (9.471)	1.815 *** (0.586)	−18.980 *** (2.277)	−33.213 * (18.397)
FE	YES	YES	YES	YES	YES	YES	YES
R ²	0.844	0.287	0.178	0.939	0.936		0.845

This study conducts a series of robustness checks to validate the regression results. First, an instrumental variable (IV) strategy is employed. The identification is based on the premise that countries located near major maritime chokepoints are more likely to be exposed to geopolitical risk [53], while geographic characteristics are largely exogenous to economic outcomes and are unlikely to directly affect SSCR directly. Accordingly, the distance from a country to the nearest maritime chokepoint is used as the instrumental variable, and a time trend term is incorporated to refine the instrument. The results indicate that the instrument passes the under-identification test, the weak instrument test, and the over-identification test, confirming its validity. The corresponding estimation results are reported in Column (3) of Table 2. Second, to mitigate potential reverse causality, both the explanatory variable and control variables are lagged by one period. The regression results are presented in Column (4) of Table 2. Third, the baseline explanatory variable is replaced with a shipping-specific geopolitical risk index, constructed following the approach of Caldara and Iacoviello [4], with the results reported in Column (5) of Table 2. Fourth, considering that panel data may suffer from autocorrelation and heteroskedasticity, the feasible generalized least squares (FGLS) estimator is employed, and the results are reported in Column (6) of Table 2. Fifth, additional control variables are incorporated. Specifically, the level of frontier technological development enhances supply chain visibility and substitution capacity, thereby affecting the buffering efficiency of maritime transport systems against external shocks. Trade openness reflects a country's dependence on maritime corridors and shapes both its exposure to risks and its capacity for risk diversification. Gross national income (GNI) per capita, as a comprehensive indicator of economic development, shapes the foundational conditions of resilience, including infrastructure quality and institutional efficiency. The corresponding regression results are presented in Column (7) of Table 2. Overall, the robustness checks consistently support the stability and reliability of the main findings.

In addition, given that geopolitical risk can manifest as observable real-world events, we further employ “battle-related deaths” and “total armed forces personnel” as proxy variables for war risk in place of the explanatory variable in the regression analysis. The estimated coefficients for these two proxies are −4.945 and −4.484, respectively, both

statistically significant at the 5% level, thereby providing further evidence of the robustness of the main results.

4.2. Heterogeneity Analysis Results

4.2.1. Industry-Level Heterogeneity Analysis

To further examine the heterogeneous effects of geopolitical risk on SSCR across industries, this study constructs industry-specific geopolitical risk indices by introducing sectoral identifiers for the energy, food, and high-tech manufacturing industries, respectively. These indicators are then separately included in the regression, and the results are reported in Table 3. The findings indicate that geopolitical risk in the energy and food sectors exerts a statistically significant negative effect on SSCR. In contrast, geopolitical risk in the high-tech manufacturing sector does not exhibit a statistically significant impact on SSCR.

Table 3. Industry-level heterogeneity analysis. * and ** denote statistical significance at the 10% and 5% levels, respectively.

	(1)	(2)	(3)
	Energy	Food	High-Tech Manufacturing
SGPR	−0.092 *	−0.122 **	−0.200
	(0.048)	(0.050)	(0.143)
Controls	YES	YES	YES
Constant	−30.493 **	−12.424	−11.542
	(11.604)	(22.845)	(19.801)
FE	YES	YES	YES
R ²	0.840	0.844	0.841

This divergence can be attributed to the fact that energy and food, as essential bulk commodities, are highly dependent on maritime corridors, characterized by long-haul shipping routes and limited substitution possibilities. When geopolitical tensions intensify in these sectors—for instance, oil tanker detentions in the Strait of Hormuz or disruptions to the Black Sea Grain Initiative—the resulting shocks are rapidly transmitted along energy- and food-related shipping supply chains. These shocks typically manifest as port blockades, shipping lane disruptions, or capacity constraints, thereby substantially weakening SSCR. By contrast, high-tech manufacturing products, such as electronics and precision instruments, are characterized by their high unit value and strong time sensitivity. Their transportation relies more heavily on air freight, implying a relatively lower dependence on shipping networks. As a result, geopolitical risk targeting the high-tech manufacturing sector—such as U.S. export controls on semiconductors—has a more limited impact on SSCR.

4.2.2. Country-Level Heterogeneity Analysis

Due to significant differences in institutional environments across countries, corruption control capacity—an important indicator of governance quality and institutional enforcement—may critically shape both the transmission channels and the intensity of geopolitical risk effects on SSCR. Accordingly, drawing on the World Bank's Corruption Control Index (percentile rankings), this study divides the sample countries into two groups, with high corruption control capacity (top 50%) and low corruption control capacity (bottom 50%), in order to examine the heterogeneous effects of geopolitical risk on SSCR. The results reported in Table 4 indicate that geopolitical risk exerts a significantly negative effect in both groups; however, the absolute magnitude of the coefficient is smaller for countries with higher corruption control capacity. This finding provides preliminary evidence that the impact of geopolitical risk on SSCR varies across countries with different institutional

quality levels. A stronger corruption control capacity helps buffer the adverse effects of geopolitical risk (as reflected in smaller coefficient estimates). This may be attributed to the fact that a well-functioning corruption control system contributes to the establishment of transparent and efficient policy and regulatory frameworks, curbs rent-seeking behavior that may distort supply chain resilience, and thereby effectively mitigates the adverse effects of geopolitical risk on SSCR.

Table 4. Country-level heterogeneity analysis. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Top 50%			Panel B: Bottom 50%		
	Full Sample	Pre-COVID-19	Post-COVID-19	Full Sample	Pre-COVID-19	Post-COVID-19
SGPR	−0.329 *** (0.110)	−1.060 ** (0.425)	−1.032 (1.105)	−1.135 * (0.568)	−2.880 * (1.460)	−0.813 (2.401)
Controls	YES	YES	YES	YES	YES	YES
Constant	−54.020 *** (12.644)	−220.282 *** (76.318)	21.962 (98.303)	−5.550 (38.377)	−430.622 ** (189.486)	21.489 (234.337)
FE	YES	YES	YES	YES	YES	YES
R ²	0.918	0.935	0.991	0.846	0.819	0.993

As a global shock, the COVID-19 pandemic has reshaped the operational structure of international shipping supply chains. This study further divides the sub-sample into two periods: pre-COVID-19 and post-COVID-19. The results in Table 4 show that the effect of geopolitical risk on SSCR becomes statistically insignificant in the post-COVID-19 period. This may be explained by the structural transformation of global supply chains. After COVID-19, global supply chains have increasingly shifted toward nearshoring and regionalization, and both the likelihood and systemic impact of single-point disruptions have been reduced. In addition, countries have strengthened self-sufficiency in critical supply chains and expanded emergency reserve capacities in the post-COVID-19 period, collectively contributing to the weakened and statistically insignificant effect of geopolitical risk on SSCR.

4.3. Mechanism Analysis

To empirically verify the proposed hypotheses, the following econometric specification is estimated:

$$M_{it} = \gamma_1 SGPR_t + \gamma_2 Z_{it} + \gamma_0 + \delta + \varepsilon_{it} \quad (2)$$

where M_{it} denotes the mechanism variables, including logistics infrastructure (*Infrastructure*), shipping freight rate volatility (*Freight*), and customs clearance efficiency (*Clearance*). Logistics infrastructure is proxied by the transport-related infrastructure quality index published by the World Bank. Shipping freight rate volatility is constructed following the volatility measurement approach of John et al. [54], using the Clarkson Container Freight Index with a three-year rolling window. Customs clearance efficiency is measured by the World Bank's clearance efficiency index. Other variables and model specifications remain consistent with Equation (1).

The results reported in Table 5 show that geopolitical risk exerts a significantly negative effect on logistics infrastructure and customs clearance efficiency, while having a significantly positive effect on shipping freight rate volatility. These findings indicate that rising geopolitical risk disrupts logistics infrastructure, amplifies shipping cost uncertainty, and weakens customs clearance performance. This is consistent with the observed effects of geopolitical risk on the mechanism variables in real-world settings. For instance, the

Russia–Ukraine conflict led to the suspension of operations at Ukrainian ports and significant damage to port infrastructure, thereby disrupting global energy and food supply chains [1]. At the same time, global fuel prices increased by approximately 64% in early 2022, which in turn raised fuel surcharges by around 50%, resulting in substantial volatility in shipping freight rates [3]. In December 2023, the spillover effects of the Israel–Palestine conflict triggered severe disruptions to global shipping routes. Approximately 80% of container vessels operating along the Red Sea–Suez Canal corridor were forced to reroute. Maersk also announced the suspension of services in the region, while informing customers of longer transit times and higher freight rates. Shipments from East Asia to the U.S. East Coast were subject to an additional surcharge of USD 1000 per TEU. In addition, geopolitical conflicts have intensified border control measures and heightened institutional uncertainty, thereby complicating cross-border logistics procedures. The hub-and-spoke transmission characteristics of port clusters further amplify these effects, leading to a substantial decline in customs clearance efficiency [43,55].

Table 5. Mechanism test results. * and *** denote statistical significance at the 10% and 1% levels, respectively.

	(1)	(2)	(3)
	Infrastructure	Freight	Clearance
SGPR	−5.300 *** (0.038)	0.080 *** (0.028)	−0.077 * (0.042)
Controls	YES	YES	YES
Constant	−217.952 *** (6.178)	−8.467 *** (2.184)	0.240 *** (0.040)
FE	YES	YES	YES
R ²	0.209	0.143	0.767

Existing studies suggest that logistics infrastructure disruption lengthens supply chain response times and reduces backup capacity, thereby constraining the ability of shipping supply chains to cope with risks and market fluctuations [31]. Increased shipping freight rate volatility further complicates cost forecasting and transport planning, reducing shipping firms’ capacity to mobilize resources during crises [36]. Moreover, declining customs clearance efficiency exacerbates information asymmetries between upstream and downstream firms, impairing cross-stage coordination within shipping supply chains [5,37]. Collectively, these mechanisms contribute to a deterioration in SSCR, thereby validating the hypotheses proposed above.

4.4. Extended Analysis Focused on the Security of Logistics Nodes

Logistics nodes play a dual role in shipping supply chains, functioning both as strategic hubs and as systemic bottlenecks [26,28]. Building on this perspective, this study further focuses on three major strategic maritime corridors—the Pacific route, the Indian Ocean route, and the Atlantic route—to identify the impact mechanisms of geopolitical risk at critical logistics nodes on SSCR. The empirical results reveal significant heterogeneity in both the effects and transmission mechanisms of geopolitical risk across different nodes (Table 6).

Table 6. Results for critical logistics nodes. The table reports the regression results after controlling for covariates and fixed effects, examining the impact of geopolitical risk on SSCR and the mechanisms, including logistics infrastructure (*Infrastructure*), shipping freight rate volatility (*Freight*), and customs clearance efficiency (*Clearance*). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A			Panel B			Panel C		
Pacific Route Nodes			Atlantic Route Nodes			Indian Ocean Route Nodes		
Panama Canal	SSCR	0.135 ** (0.055)	Black Sea Route	SSCR	−0.063 ** (0.026)	Red Sea Route	SSCR	−1.064 ** (0.435)
	Infrastructure	0.935 *** (0.007)		Infrastructure	−0.435 *** (0.003)		Infrastructure	−7.377 *** (0.053)
	Freight	2.280 *** (0.167)		Freight	6.271 *** (0.106)		Freight	15.643 *** (0.771)
	Clearance	−0.044 * (0.024)		Clearance	−0.077 * (0.042)		Clearance	−0.177 * (0.097)
Strait of Malacca	SSCR	0.656 ** (0.268)	Cape of Good Hope	SSCR	−0.014 (0.034)	Port of Djibouti	SSCR	−0.256 ** (0.105)
	Infrastructure	4.549 *** (0.032)		Infrastructure	−0.057 (0.063)		Infrastructure	−1.778 *** (0.013)
	Freight	0.105 (0.083)		Freight	0.368 (0.454)		Freight	3.811 *** (0.092)
	Clearance	−0.221 * (0.121)		Clearance	−0.004 (0.006)		Clearance	−0.021 * (0.012)
Port of Shanghai	SSCR	−0.059 ** (0.024)	Port of New York and New Jersey	SSCR	−0.210 ** (0.086)	Port of Jeddah	SSCR	−0.412 ** (0.169)
	Infrastructure	−0.410 *** (0.003)		Infrastructure	−1.456 *** (0.010)		Infrastructure	−2.858 *** (0.020)
	Freight	1.902 *** (0.166)		Freight	1.961 *** (0.049)		Freight	0.913 *** (0.074)
	Clearance	−0.589 * (0.323)		Clearance	−0.253 * (0.138)		Clearance	−0.118 * (0.065)
Port of Los Angeles	SSCR	−0.236 ** (0.096)	Port of Rotterdam	SSCR	−0.097 ** (0.040)	Port of Colombo	SSCR	−0.028 (0.040)
	Infrastructure	−1.634 *** (0.012)		Infrastructure	−0.674 *** (0.005)		Infrastructure	−0.081 (0.068)
	Freight	1.676 *** (0.228)		Freight	0.273 (0.188)		Freight	0.168 ** (0.065)
	Clearance	−0.126 * (0.069)		Clearance	−0.177 * (0.097)		Clearance	0.005 (0.011)

The main findings are summarized as follows:

1. Geopolitical risk at most critical nodes exerts a significantly negative impact on SSCR.

Across the majority of critical logistics nodes, the estimated coefficients of geopolitical risk are significantly negative, indicating that geopolitical conflicts occurring at these nodes substantially weaken SSCR. Notably, however, geopolitical risk at the Panama Canal and the Strait of Malacca exhibits a significantly positive coefficient. This may be explained by the fact that these two nodes are among the very few irreplaceable chokepoints in the global maritime transport system. When geopolitical risk emerges at such critical nodes, the willingness and capacity of the international community and major powers to intervene are considerably stronger than at ordinary nodes. At the same time, shipping companies tend to proactively adjust voyage schedules, increase strategic stockpiles of critical supplies, and activate contingency plans for alternative capacity. As a result, stricter transit safeguards and enhanced security coordination are temporarily established, which in turn improve SSCR in the short term.

2. The mechanism of logistics infrastructure disruption is particularly pronounced in ports.

Further analysis indicates that, among the twelve critical logistics nodes examined in this study, geopolitical risk at eight nodes reduces SSCR through the disruption of logistics infrastructure, with this effect being primarily concentrated in port-type nodes. This may be attributed to the fact that ports themselves constitute the physical embodiment of logistics infrastructure. When geopolitical risk originates from port-type nodes, conflicts directly affect port facilities, where cranes, warehousing systems, and berthing infrastructure may suffer direct physical damage or functional disruption. In contrast, for strait- and canal-type nodes, geopolitical risk is more often driven by surrounding regional conflicts or navigational security threats. The spatial distance between the risk source and their infrastructure is relatively greater, resulting in a limited impact on their physical infrastructure. In these cases, the effects for strait- and canal-type nodes are more likely to manifest as transit restrictions, security-related delays, or route deviations, rather than direct physical damage to infrastructure.

3. Logistics nodes along the Indian Ocean route exhibit more pronounced effects on freight rate volatility mechanisms.

Compared with the Pacific and Atlantic routes, the coefficients of freight rate volatility at critical logistics nodes along the Indian Ocean route are consistently highly significant. This phenomenon can be attributed to the relatively inelastic supply structure of the Indian Ocean route, where alternative pathways are limited. Key nodes such as the Red Sea corridor, the Port of Djibouti, and the Port of Jeddah are closely connected to the Suez Canal route and jointly constitute a critical maritime corridor between Asia and Europe, occupying a strategic position within the global shipping network. Once geopolitical risk escalates, shipping firms face a lack of cost-effective alternative routes. This low supply elasticity renders freight rates highly sensitive to risk shocks, thereby significantly amplifying volatility.

4. The reduced customs clearance efficiency represents a common transmission channel across most nodes.

Across major logistics nodes, the coefficients of customs clearance efficiency are predominantly significantly negative. This suggests that the reduced customs clearance efficiency serves as a common transmission channel through which geopolitical risk affects SSCR. In response to geopolitical tensions, countries may frequently adjust tariffs, quotas, and embargo measures, or even revise relevant legal frameworks. Such institutional changes directly affect customs clearance procedures, leading to higher compliance costs, increased procedural complexity, and reduced clearance efficiency. As customs clearance functions as a critical institutional gateway for all cross-border maritime cargo, its operational efficiency is highly sensitive to institutional shocks. Therefore, reduced customs clearance efficiency becomes a common transmission channel.

5. Discussion

Against the backdrop of continuously escalating geopolitical risk and its persistent disruption to global shipping networks, this study provides an empirical response to Notteboom et al. [5], who examine the impact of major shock events on the stability of port and shipping systems. Consistent with their findings, we confirm that geopolitical risk significantly reduces SSCR. In addition, Su et al. [28] argue that the negative impact of geopolitical risk on SSCR is shaped by adjustments in supply chain configuration. Our study further finds that the effect of geopolitical risk on SSCR becomes statistically insignificant in the post-COVID-19 period, when global supply chains have shifted toward

nearshoring and regionalization. This provides new empirical evidence for understanding the complex relationship between geopolitical risk and SSCR in the post-COVID-19 era.

This study further examines the transmission mechanisms through which geopolitical risk affects SSCR. The results show that geopolitical risk undermines SSCR by disrupting logistics infrastructure. This finding not only corroborates Kashav et al. [21], who identify infrastructure disruption as the most severe consequence of geopolitical risk, but also aligns with Tsoulfas [31], who emphasizes that physical damage leads to infrastructure disruption and maintenance difficulties, thereby reducing supply chain flexibility and adaptability. In addition, Ju et al. [36] find that the intensified freight rate volatility since 2013 has had adverse effects on both shippers and ports. This study also incorporates freight rate volatility into the analytical framework of the relationship between geopolitical risk and SSCR, highlighting its role as a key transmission channel of risk. Finally, this study finds that geopolitical risk undermines SSCR by reducing customs clearance efficiency. This is consistent with Notteboom et al. [33] and Pishchulov et al. [37], who show that trade policy uncertainty hampers supply chain efficiency. It further enriches the literature on how external shocks affect supply chain operations and resilience.

In the extended analysis, this study focuses on three major maritime corridors—the Pacific Route, the Indian Ocean Route, and the Atlantic Route—and systematically identifies the impact of geopolitical risk at critical logistics nodes on SSCR. The results reveal significant heterogeneity in the effects of geopolitical risk across different nodes, with transmission mechanisms exhibiting distinct heterogeneous characteristics. This finding is consistent with Notteboom et al. [5], who argue that the impacts of exogenous shocks on shipping supply chains vary across regions and ports. This study shows that geopolitical risk has a significantly negative effect on SSCR at most nodes. This is primarily because these nodes are simultaneously exposed to multiple pressures, including fragile infrastructure, constrained customs clearance processes, and heightened market expectations and price volatility, making them complex carriers of risk transmission. This perspective is consistent with Su et al. [28] and further validates the findings of Yue et al. [26], who demonstrate that once existing maritime corridors are disrupted, cascading effects such as cargo congestion and capacity mismatch may be triggered, leading to systemic congestion across global transport networks.

5.1. Academic Contributions

First, this study extends the research boundary of supply chain resilience theory. Existing studies on manufacturing and general supply chain resilience primarily emphasize production- and organization-based resilience within value-adding processes, focusing on the adaptability and recovery capacity of production systems [13]. However, as Lam and Bai [14] point out, the essence of shipping supply chains lies in the efficient movement of goods across spatial and temporal dimensions, implying a fundamentally different resilience logic from that of manufacturing systems. This study constructs a three-dimensional evaluation framework for SSCR encompassing function, structure, and governance. The theoretical contribution lies in expanding resilience from the concept of “recovery capability” to a dynamic triadic capacity that integrates functional maintenance, structural optimization, and governance adaptability. This complements the existing literature and provides a solid scientific basis for developing a proactive and adaptive resilience framework for the shipping industry.

Second, this study advances the measurement of geopolitical risk from a generic framework toward an industry-adapted analytical tool, thereby enhancing both the theoretical precision and contextual applicability of the construct. In contrast to conventional approaches that conceptualize geopolitical risk as an exogenous and homogeneous shock,

this paper emphasizes risk perception and applies the general geopolitical risk framework developed by Caldara and Iacoviello [4] to the shipping industry. This approach is consistent with the recent work of Bondarenko et al. [32], who apply the same framework to the analysis of economic sanctions against Russia. From a maritime transport system perspective, this refinement further demonstrates that geopolitical risk is not a uniform external shock; rather, its transmission channels and impact intensity depend on the structural characteristics and operational logic of the shipping industry. Accordingly, this study offers a transferable analytical framework for future research on industry-specific manifestations of geopolitical risk.

Third, this study advances the spatial and hierarchical understanding of supply chain resilience. Existing research has shown that, despite exposure to exogenous shocks, global supply chain networks exhibit a certain degree of robustness, yet remain highly vulnerable to targeted disruptions at critical logistical nodes [26]. Building on this insight, this study conducts an extended analysis to systematically identify the differentiated impacts of geopolitical risk at critical logistics nodes—such as the Strait of Malacca, the Black Sea routes, and the Red Sea corridor—on SSCR. López et al. [27] note that the literature has paid insufficient attention to the heterogeneous responses of different industries, supply chain tiers, and geographic regions under geopolitical risk. This study directly addresses this gap, providing systematic empirical evidence from the perspective of risk transmission mechanisms at critical logistics nodes. In doing so, it offers a novel analytical lens for designing shipping supply chain risk-containment mechanisms based on the identification of key logistical chokepoints.

5.2. Policy Implications

1. A multidimensional and coordinated resilience framework should be established.

To effectively respond to the complex disruptions induced by geopolitical risk, policy-makers need to act jointly across three dimensions: functional, structural, and governance. At the functional level, greater diversification of both origin and destination markets within shipping supply chains should be promoted. This enhances supply-side competitiveness and reduces systemic vulnerability arising from disruptions at single nodes. As emphasized by Su et al. [28], persistent geopolitical risk calls for stronger international cooperation and the development of resilient ecosystems capable of rapid adaptation. At the structural level, greater emphasis should be placed on monitoring and optimizing liner shipping connectivity. Multi-source data, particularly from Automatic Identification Systems (AISs), can be integrated to dynamically assess vulnerabilities in shipping networks [44]. In parallel, the digital transformation of the shipping sector should be accelerated. The transformative potential of data analytics, artificial intelligence, the Internet of Things, and cloud computing should be fully leveraged [40]. At the governance level, efforts should focus on enhancing port throughput capacity and improving the efficiency of resource allocation, thereby strengthening the position within the global production network. At the same time, policies should respond to core customer demands, including cost competitiveness, environmental sustainability, resource efficiency, and safety assurance. The adoption of key technologies, such as green vessel design and high-efficiency engines, should be promoted [56].

2. Differentiated risk governance should be implemented across sectors, countries, and time horizons.

Industry-level heterogeneity indicates that the impact of geopolitical risk on SSCR varies with the degree of dependence on shipping routes. For essential bulk commodities such as energy and food, which rely heavily on maritime corridors, governments should incorporate them into national strategic supply systems. Dedicated shipping corridor protection mechanisms should be established, alongside strategic stockpiling to buffer

against logistics disruptions. For high-technology manufacturing sectors, particular attention should be given to critical raw materials sourced from highly concentrated suppliers. National-level strategic reserve systems should be developed to reduce the risk of supply interruptions arising from technological restrictions or the blockade of specific nodes. Cross-country heterogeneity shows that the negative impact of geopolitical risk on SSCR is smaller in magnitude, yet remains highly statistically significant in countries with stronger corruption control. For these countries, improving risk governance is not sufficient. More responsive risk early warning systems should be established, with a particular focus on the dedicated monitoring of geopolitical risk. This helps prevent the amplification of shocks due to high transmission efficiency. Temporal heterogeneity analysis further indicates that the impact of geopolitical risk on SSCR becomes insignificant in the post-COVID-19 period. This suggests that greater efforts should be made to strengthen mutual recognition arrangements and cooperation mechanisms with major trading partners in customs clearance and inspection standards. Such efforts would help maintain the smooth and efficient functioning of shipping supply chains amid the post-COVID-19 reconfiguration of global supply networks.

3. Targeted interventions should focus on key transmission channels.

Mechanism analysis identifies three primary channels through which geopolitical risk undermines SSCR. First, with respect to logistics infrastructure disruption, protection and redundancy design for critical assets—such as ports and shipping lanes—should be strengthened, and rapid post-disruption recovery capabilities should also be enhanced [21,31]. Second, to address freight rate volatility, mechanisms aimed at stabilizing rate fluctuations or hedging instruments should be developed to reduce the impact of sharp price swings on supply chain stability [36]. Third, in response to declining customs clearance efficiency, efforts should be made to promote the standardization and digitalization of international clearance procedures, thereby reducing compliance costs and uncertainty arising from trade restrictions and regulatory adjustments [33,37].

4. Differentiated risk management should be implemented at critical logistics nodes.

It is necessary to establish a tiered risk management system based on node importance and vulnerability. On the one hand, key maritime corridors should be subject to enhanced monitoring and redundancy design. This includes diversified routing strategies, backup port agreements, and emergency capacity reserves, all of which can reduce the systemic impact of disruptions at single nodes on the global network. On the other hand, differentiated response strategies should be implemented based on the specific risk sources faced by each node. When infrastructure damage is the primary risk, port modernization and digital upgrading should be advanced. When customs clearance efficiency declines, international coordination should be strengthened to promote the standardization of clearance procedures. When freight rate volatility emerges, information-sharing platforms can be used to stabilize market expectations and prevent systemic congestion triggered by panic-driven capacity reallocation [26].

6. Conclusions

This study employs a panel dataset of 42 coastal countries over the period 2007–2023 to systematically examine the impact of geopolitical risk on SSCR and its underlying mechanisms. The results are organized into three main findings. First, geopolitical risk significantly reduces SSCR. Industry-level heterogeneity analysis shows that geopolitical risk in the energy and food sectors exerts a significant negative effect on SSCR, whereas in high-technology manufacturing it is statistically insignificant. Country-level heterogeneity further reveals that stronger corruption control capacity helps buffer the adverse effects

of geopolitical risk. Moreover, in the post-COVID-19 period, the impact of geopolitical risk on SSCR becomes statistically insignificant. Second, mechanism analysis suggests that geopolitical risk undermines SSCR primarily through three channels: logistics infrastructure disruption, increased freight rate volatility, and reduced customs clearance efficiency. Third, evidence from critical logistics nodes further shows that the impact of geopolitical risk at most nodes on SSCR is significantly negative. The mechanism of logistics infrastructure disruption is particularly pronounced in ports. Logistics nodes along the Indian Ocean route exhibit more pronounced effects on freight rate volatility channel, while reduced customs clearance efficiency represents a common transmission channel across most nodes.

Based on the findings, this study proposes four policy implications. First, a three-dimensional resilience framework should be established across functional, structural, and governance dimensions. This calls for supply chain diversification, strengthened liner connectivity monitoring, accelerated digital transformation, and the wider adoption of green shipping technologies to enhance systemic resilience and support sustainable maritime development. Second, risk governance should be differentiated across sectors, countries, and time horizons. Energy and food security should be embedded in national strategic systems. Countries with stronger corruption control require more responsive and targeted early warning mechanisms. In the post-COVID-19 context, mutual recognition of standards with major trading partners should be strengthened to ensure supply chain continuity and efficiency. Third, policy interventions should target key transmission channels. Priority should be given to strengthening port and shipping lane resilience through redundancy design and post-disruption recovery capacity. Freight rate volatility should be mitigated through stabilization mechanisms or hedging instruments. Meanwhile, customs procedures should be streamlined and digitalized to improve clearance efficiency. Fourth, differentiated risk management should be implemented at critical logistics nodes. Key maritime corridors require enhanced monitoring, redundant routing, backup port agreements, and emergency capacity reserves to reduce vulnerability to single-node disruptions. In addition, infrastructure modernization, customs standardization, and information-sharing platforms should be deployed according to specific risk sources.

This study has several limitations. The existing literature typically defines supply chain resilience as firms' post-shock recovery performance and relies on transaction- or firm-level high-frequency data for measurement. However, such data are difficult to obtain in cross-border shipping contexts. Constrained by data availability, this study adopts a macro-level perspective and constructs proxy indicators of SSCR using country-level data across functional, structural, and governance dimensions. Accordingly, the "resilience" measured here mainly reflects countries' ex ante risk mitigation and adaptive capacity under geopolitical risks, as captured by policy adjustments, resource reallocation, and network reconfiguration, rather than ex post firm-level recovery. This inevitably limits the direct characterization of recovery processes and the identification of heterogeneous firm responses to geopolitical risk. In addition, as digital technologies increasingly permeate maritime transport, available statistical and textual data remain limited in capturing digitalization and its role in shaping resilience. Future research could enhance data granularity by using firm-level or shipping-order-level high-frequency data. This would allow for a deeper examination of how digital technologies, through improving information transparency, optimizing cross-node coordination, and strengthening real-time decision-making, influence SSCR at the micro level, thereby enabling a multi-level analysis linking macro and micro resilience dynamics.

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Appendix A

From a functional perspective, manufacturing supply chain resilience is primarily value oriented and emerges from internal transformation processes such as raw material conversion and production optimization [13]. By contrast, SSCR centers on the reliability and continuity of international cargo flows. It emphasizes cross-border spatiotemporal coordination and prioritizes transport timeliness, cost efficiency, and network stability [14]. From a structural perspective, manufacturing and other generic supply chains are often shaped by intelligent production systems and full life-cycle product management. As a result, they tend to rely on regionally clustered resilience networks [15]. Shipping supply chains, however, require global network coordination and logistics process transparency. They therefore evolve toward hub-and-spoke-oriented network structures, supported by maritime digital technologies and anchored in critical logistical nodes such as hub ports and maritime chokepoints [12,16]. From a governance perspective, risks in manufacturing and other generic supply chains are typically associated with short-cycle fluctuations and technological rigidity. Correspondingly, resilience strategies often emphasize modular design, nearshoring, or outsourcing [17,18]. Shipping supply chains operate under different risk conditions. Their risks usually have a long cycle, are path dependent, and are geographically rigid, with disruptions propagating across borders and regions. As a result, resilience governance in shipping relies on alternative ports and routes, redundancy in critical resources, and network-level capacity building to mitigate transport delays and network disruptions [5,19].

Appendix B

Table A1. Summary of methods for SSCR.

Reference	Method	Brief Characterization	Limitations
Xu et al. (2022) [11]	Quantitative	Developing a nonlinear capacity-based cascading failure model, integrating shipping connectivity and port load data to simulate load redistribution under disruptions and assess dynamic network vulnerability.	Assumption-based parameter settings (capacity and redundancy); neglect of time delays in load adjustment; discrepancies between theoretical and actual capacity levels.
Liu et al. (2023) [12]	Mixed Method	Constructing a three-layer resilience framework (goals–strategies–practices). Integrating AHP (weighting), QFD (relationship mapping), and DEMATEL (causal analysis) based on expert survey data to identify key resilience drivers.	Strong dependence on expert scoring; limited regional coverage; lack of dynamic and behavioral analysis.

Table A1. *Cont.*

Reference	Method	Brief Characterization	Limitations
Dui et al. (2021) [20]	Quantitative	Proposing a new method to optimize the residual resilience management of ports and routes in MTS and developing an optimal resilience model. Employing the Copeland method to rank the importance of ports and routes.	Lack of differentiation between types of disruptions; incomplete incorporation of recovery and redundancy costs.
Kashav et al. (2022) [21]	Mixed Method	Identifying 46 resilience barriers through a literature review and Delphi survey. Applying fuzzy AHP to evaluate their relative importance and conducting sensitivity analysis to ensure robustness.	Case specific (limited to a single country); heavily reliant on expert judgment; neglects interdependencies among factors.
Karakas et al. (2024) [22]	Mixed Method	Developing a multi-criteria resilience assessment framework incorporating indicators such as density, demand, and diversity, combining objective and subjective MCDM approaches.	Strong case dependence; subjective weighting of indicators.
Wu et al. (2024) [23]	Quantitative	Based on the global container shipping network (GCSN), identifying collapse thresholds and measures attack tolerance (i.e., the maximum proportion of disrupted ports before network failure). Tracking its temporal evolution (two-year window) to assess vulnerability dynamics.	Focus on intentional attack scenarios; limited capture of adaptive resilience in multi-hub network structures; limited consideration of node heterogeneity.

Appendix C

Table A2. News keyword retrieval structure for measuring geopolitical risk indicators. Bold text indicates category labels.

Search scope	Full text				
Language	Chinese				
Search period	2007–2023				
Database	China National Knowledge Infrastructure (CNKI)				
Newspapers	People’s Daily, Guangming Daily, Economic Daily, Financial Times, Economic Information Daily, China Securities Journal, People’s Daily Overseas Edition, International Business Daily, China Economic Times, China Water Transport, China Transport News, China Trade News, Modern Logistics News, Securities Daily				
Keywords	Geopolitical risk phrases				
	<table> <tr> <th>Theme</th><th>Actions</th></tr> <tr> <td>Threats to peace</td><td>Threat; Warning; Fear; Risk; Danger; Alert; Suspicion; Crisis; Trouble; Dispute; Tension; Urgency; Inevitability; Brinkmanship; Edge; Intimidation; Rejection; Boycott; Sabotage</td></tr> </table>	Theme	Actions	Threats to peace	Threat; Warning; Fear; Risk; Danger; Alert; Suspicion; Crisis; Trouble; Dispute; Tension; Urgency; Inevitability; Brinkmanship; Edge; Intimidation; Rejection; Boycott; Sabotage
Theme	Actions				
Threats to peace	Threat; Warning; Fear; Risk; Danger; Alert; Suspicion; Crisis; Trouble; Dispute; Tension; Urgency; Inevitability; Brinkmanship; Edge; Intimidation; Rejection; Boycott; Sabotage				

Table A2. Cont.

War-related activities	War; Conflict; Hostilities; Revolution; Uprising; Rebellion; Coup; Geopolitics	Start; Declaration; Outbreak; Breakthrough; Launch; Initiation
Military activities	Military; Troops; Missiles; Weapons; Bombs; Warheads; Alliance; Enemy; Insurgency; Army; Navy; Air Force; Armed Forces; Russia	Mobilization; Sanctions; Blockade; Embargo; Quarantine; Ultimatum; Deployment; Advance; Attack; Offensive; Expulsion; Shelling; Invasion; Clash; Strike; Launch
Piracy and terrorism	Terrorism; Guerrilla; Hostage; Piracy; Somalia	Attack; Operation; Explosion; Killing; Strike; Hijacking; Assault; Gulf of Aden; Pirate Attacks; Maritime Search and Rescue
Shipping-specific phrases		
	Theme	Actions
Industry-level shipping activities	Maritime Industry; Shipping Industry; Water Transport; Maritime Authority; Maritime Administration; Shipping Market; Port and Shipping Sector; Liner Conference; Maritime Regulation	Maritime Transport; Shipping; Waterborne Transport; Sea Transport; Inland Waterway Transport; Coastal Transport; Sea–Rail Intermodal Transport; Freight Rates; Shipping Prices; Shipping Routes; International Maritime Regulations; Shipping Agreements; Route Tariffs; Supply Chains; Spot Market Rates; Cross-Strait Direct Shipping; Pandemic Prevention and Control
Shipping firms and operations	Shipping Companies; Liner Companies; Maritime Enterprises; Ocean-Going Shipping; Maersk; COSCO; COSCO Group; Port and Shipping Enterprises	Cargo Transport; Container Transport; Bulk Cargo Transport; LNG Transport; Energy Transport; Oil Transport
Port and shipping infrastructure	Dry Bulk Carriers; Container Ships; Containers; TEU; Terminals; Ports; Vessels; Tianjin Port	Throughput; Standard Containers; Dry Bulk; Belt and Road Initiative; Yangtze River Shipping; Smart Shipping; Pinglu Canal; Beibu Gulf

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